Handling rescue medication in diabetes trials

Comparing standard and causal inference methods

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Louise Jespersen

Researcher at Institute of Medical Statistics, Center for Medical Data Science, Medical University of Vienna

louise.jespersen@meduniwien.ac.at

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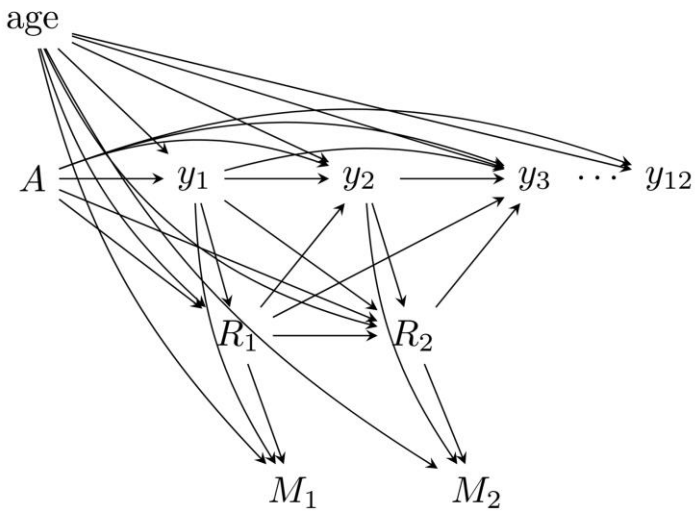
The CONFIRMS consortium consists of four institutions, Uppsala University, University Medical Center Göttingen, Austrian Agency for Health and Food Safety, and Medical University of Vienna.

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Diabetes trial with rescue medication

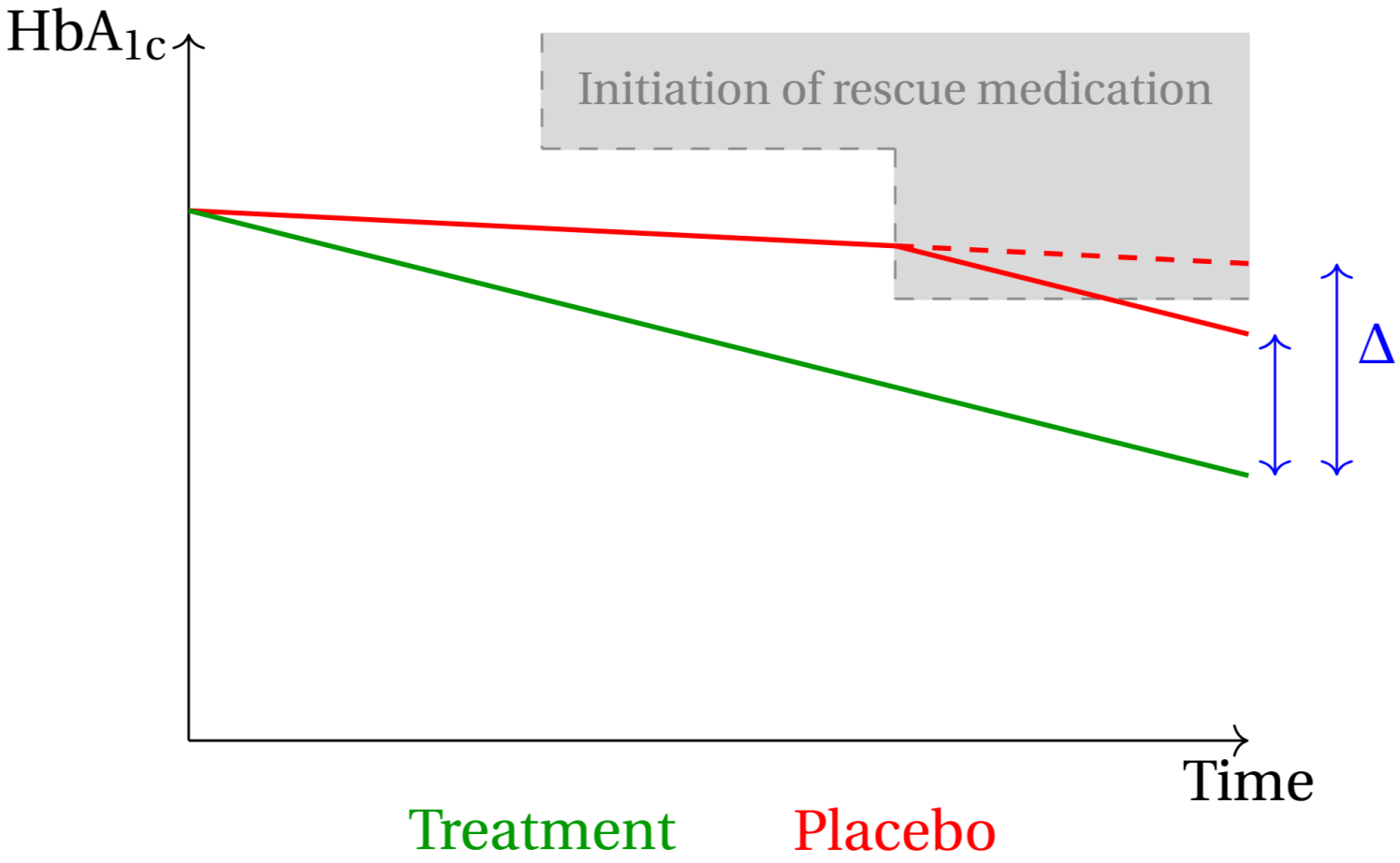
Trial setup:

- 1:1 randomisation, one year study time, monthly visits.
- Primary endpoint is change from baseline in HbA1c after one year.
- Intercurrent events: Use of rescue medication and treatment discontinuation.
- Patients might initiate rescue medication when glycaemic control is insufficient.
- Monotone missingness and rescue medication intake.



Objective

Investigate how methods, to account for initiation of rescue medication, perform.



Rescue medication defines two different treatment effects

Estimand as the clinical question:

- In both estimands, the endpoint is the mean difference in one-year change from baseline in HbA1c. Treatment discontinuation is handled using treatment policy.
  - Treatment policy strategy: Treatment effect regardless of use of rescue medication.
  - Hypothetical strategy: Treatment effect in the hypothetical scenario where rescue medication would not be made available.

Investigated scenarios

Scenario	Mechanism	Characteristics
Core scenario		≈10% rescue medication; ≈5% missing
Increased missingness	Probabilistic, conditional on covariate history.	≈13% missing; stronger HbA1c dependence
Increased rescue medication	Probabilistic, conditional on covariate history.	≈34% receive rescue medication
Positivity violation		Rescue medication largely determined by HbA1c, 2:1 ratio

Methods for treatment policy strategy

Inverse probability weighting (IPW):

- Reweights observed outcomes using probabilities of remaining observed. Explanatory variables: *A, age, HbA1c, and* R_j. ANCOVA analysis, for change from baseline in HbA1c, adjusted for baseline covariates.

Multiple imputation (MI):

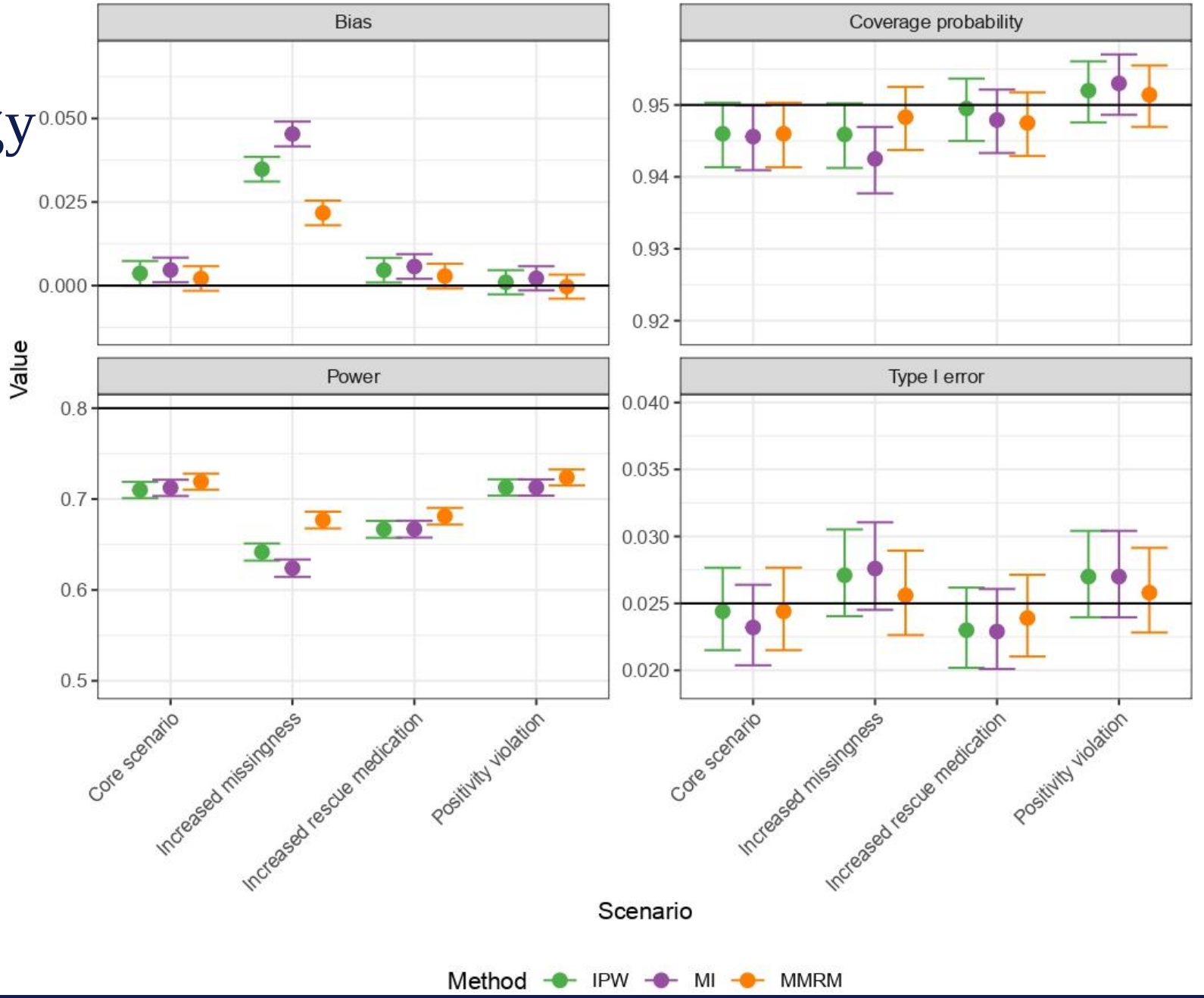
- Imputes missing longitudinal outcomes before final-visit analysis. Imputation model included: *age, HbA1c, and* R_j. ANCOVA analysis, for change from baseline in HbA1c, adjusted for baseline covariates.

Mixed model for repeated measures (MMRM):

- Models the repeated measures of change of HbA1c with unstructured covariance. Explanatory variables: *A, age, baseline HbA1c,* $visit_j$.

Treatment policy strategy

- The methods performed similarly under favourable or moderately challenging conditions.
- Greater missingness increased variability, causing more bias.



Methods for the hypothetical strategy

IPW, MI, MMRM:

- Applied as previously described, but treating post-rescue outcomes as missing.

De-mediation (DM):

- Recursive procedure applied at every visit.
- Outcomes after rescue medication are adjusted by subtracting the estimated effect of the rescue medication, $\hat{\psi}$, and the adjusted outcome values are compared between groups.
- Model based variance estimation.[1]

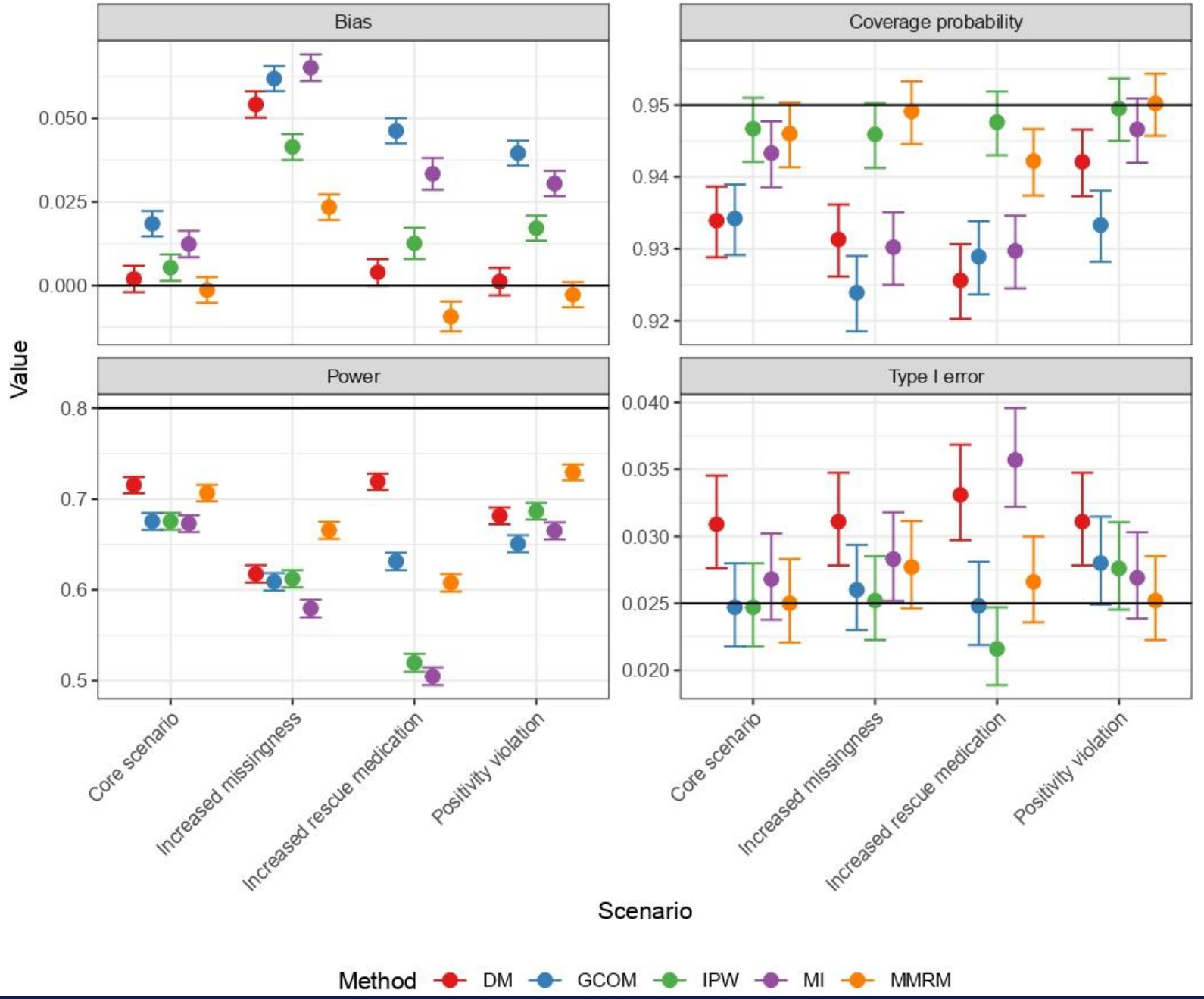
G-computation (GCOM):

- Uses fitted longitudinal models to simulate outcomes under a no-rescue intervention.

[1]Lasch & Guizzaro et. al., 2023 and 2025.

Hypothetical strategy

- MMRM and de-mediation often performed best in terms of bias and power.
- De-mediation showed an overall inflation of type I error. Bootstrap inference for de-mediation remains to be investigated.



Limitations

- Bootstrap inference for de-mediation remains to be investigated.
- In the data generating mechanism, HbA1c is the only time-varying covariate.
- Conclusions are specific to the investigated estimands, mechanisms, and model specifications.

Conclusion

Treatment policy strategy:

- Methods performed similar, no strong evidence that IPW outperforms the standard methods

Hypothetical strategy:

- Stronger dependency on modelling choices
- Stronger inflation of type I error, IPW provides the best control
- De-mediation showed less bias

Choice of methods, and sample size planning, should be made according to the expected mechanisms of the intercurrent events.

References

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Code from the project available here: https://github.com/CI-RCT-Sim/CI.RCT.Sim

Thank you for your attention

Louise Jespersen, louise.jespersen@meduniwien.ac.at

Institute of Medical Statistics, Center for Medical Data Science

Medical University of Vienna

Scan the QR-code for the documents and code of the project

Backup slides

Louise Jespersen

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Data generation

$$age \sim N(60,10), \qquad \mu_0 \sim N(8,1), \qquad \alpha_{resp,i} \sim Unif(0,1),$$
$$r_{ij} \sim N(0,\Sigma), \qquad cor(r_{ij}, r_{ij'}) = 0.5, \qquad \text{var}(r_{ij}) = 1$$

$$y_{ij} = \mu_0 + j/k \cdot s_{age,i} + \beta_{trt,j} \cdot \alpha_{resp,i} \cdot A_i + r_{ij}.$$
$$s_{age,i} = 2exp\left(-b_{age} \cdot (age_i - 30)\right)$$
$$\beta_{trt} = exp(-\log(2) \cdot j/2) - 1$$

$$log\left(\frac{p_{resc,ij}}{1 - p_{resc,ij}}\right) = \beta_{resc,0} + (y_{ij} - 10) \cdot \beta_{resc,y} + (age_i - 60) \cdot \beta_{resc,age}$$

Outcomes after rescue medication: $y_{ij} + \beta_{resc,j*} \cdot \alpha_{resp,resc,i} \cdot R_{ij}$

Details of the simulation study

- 10.000 iterations per scenario
- Hypothesis tests were performed as one-sided tests at the one-sided 2.5% significance level.
- Confidence intervals were calculated as two-sided intervals with a nominal coverage probability of 95%.
- Assessed operating characteristics
  - Bias
  - Power
  - Type I error rate
  - Confidence interval coverage